Primer: Maximum Causal Entropy IRL (MCE-IRL)

econirl package documentation

1 Problem and Motivation

Ziebart (2010) introduced Maximum Causal Entropy IRL (MCE-IRL), which conditions each action only on information available at decision time by replacing Shannon entropy $H(A^T)$ with causal entropy $H(A_t|S_{1:t}, A_{1:t-1})$. This avoids non-causal visitation calculations that can propagate future information backward. The resulting softmax Bellman equation is simultaneously the MCE optimal policy (Theorem 1 of Ziebart 2010) and the structural logit choice model of Rust (1987), making MCE-IRL the bridge between econometrics and IRL.

2 Model and Notation

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\|s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

The agent solves the maximum causal entropy problem

$$\max_\pi \; \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t; \theta) + \sigma \mathcal{H}(\pi(\cdot|s_t)) \big) \right] \tag{1}$$

subject to the feature matching constraint $\mathbb{E}_\pi[\phi] = \mathbb{E}_{\text{data}}[\phi]$, where $r(s, a; \theta) = \theta^\top \phi(s, a)$ is linear in known features ϕ with unknown weights θ.

Symbol	Name	Definition
$D_\pi(s)$	State visitation	Expected discounted visits to s under π
ρ_0	Initial distribution	Empirical start-state frequencies
F_π	Policy transitions	$F_\pi[s, s'] = \sum_a \pi(a\|s)\, P(s'\|s, a)$

3 Derivations

3.1 Theorem 1: The MCE Optimal Policy

We seek the policy that is maximally noncommittal (highest entropy) while reproducing the observed feature statistics. The *causal* entropy conditions each action only on information available at decision time, respecting the agent's sequential information structure.

Primal problem.

$$\max_{\pi} \; H_{\text{causal}}(\pi) \quad \text{s.t.} \quad \mathbb{E}_{\pi}[\phi] = \bar{\phi}_{\text{data}}, \quad \pi(a|s) \geq 0, \quad \sum_{a} \pi(a|s) = 1 \; \forall s. \tag{2}$$

The causal entropy is

$$H_{\text{causal}}(\pi) = \sum_{t=0}^{\infty} \gamma^t \sum_{s} d_{\pi}^{t}(s) \sum_{a} \pi(a|s) \log \frac{1}{\pi(a|s)}, \tag{3}$$

where $d_{\pi}^{t}(s) = P(S_t = s \mid \pi)$ is the probability of being in state s at time t under policy π.

Lagrangian. Introducing multipliers $\theta \in \mathbb{R}^K$ for the feature constraints and $\lambda(s)$ for the normalization constraints,

$$\mathcal{L}(\pi, \theta, \lambda) = H_{\text{causal}}(\pi) + \theta^{\top}\big(\mathbb{E}_{\pi}[\phi] - \bar{\phi}_{\text{data}}\big) + \sum_{s} \lambda(s) \left(\sum_{a} \pi(a|s) - 1 \right). \tag{4}$$

Substituting the definitions and taking the derivative with respect to $\pi(a|s)$,

$$\frac{\partial \mathcal{L}}{\partial \pi(a|s)} = d_{\pi}(s)\big[-\log \pi(a|s) - 1 + \theta^{\top}\phi(s,a) + \text{continuation}\big] + \lambda(s) = 0. \tag{5}$$

The "continuation" term accounts for how changing $\pi(a|s)$ affects future state visitation. Solving and absorbing constants into $\lambda(s)$, we obtain

$$\pi^*(a|s) = \frac{\exp\big(Q^*(s,a)/\sigma\big)}{\sum_{a'} \exp\big(Q^*(s,a')/\sigma\big)}, \tag{6}$$

where Q^* satisfies the soft Bellman recursion

$$Q(s,a) = \theta^{\top}\phi(s,a) + \beta \sum_{s'} P(s'|s,a)\, V(s'), \tag{7}$$

$$V(s) = \sigma \log \sum_{a} \exp\big(Q(s,a)/\sigma\big). \tag{8}$$

The policy is therefore

$$\pi(a|s) = \exp\big((Q(s,a) - V(s))/\sigma\big). \tag{9}$$

This is exactly the structural logit model of Rust (1987) with Type I extreme value errors at scale σ. The Lagrange multipliers θ play the role of the structural parameters, and the soft Bellman recursion is the standard DDC value function.

3.2 Theorem 2: Feature Matching Gradient

Dual problem. Define the dual function $G(\theta) = \max_{\pi} \mathcal{L}(\pi, \theta)$. Since the inner maximization has the closed-form solution $\pi^*(\theta)$ from Theorem 1, we can write

$$G(\theta) = H_{\text{causal}}(\pi^*(\theta)) + \theta^{\top}\big(\mathbb{E}_{\pi^*}[\phi] - \bar{\phi}_{\text{data}}\big). \tag{10}$$

The MCE-IRL problem is $\min_{\theta} G(\theta)$.

Gradient by the envelope theorem. Since π^* is the unconstrained maximizer of $\mathcal{L}$ for fixed θ, the envelope theorem gives

$$\nabla_\theta G(\theta) = \mathbb{E}_{\pi^*(\theta)}[\phi] - \bar{\phi}_{\text{data}} = \bar{\phi}_\pi(\theta) - \bar{\phi}_{\text{data}}. \tag{11}$$

The expected features under the current policy are computed via the occupancy measure

$$\bar{\phi}_\pi(\theta) = \sum_s D_\pi(s) \sum_a \pi(a|s)\, \phi(s,a), \tag{12}$$

where $D_\pi(s)$ is the expected discounted state visitation frequency under π. This satisfies the fixed-point equation

$$D_\pi = \rho_0 + \beta\, F_\pi^\top\, D_\pi \quad \Longrightarrow \quad D_\pi = (I - \beta\, F_\pi^\top)^{-1}\, \rho_0, \tag{13}$$

with $F_\pi[s,s'] = \sum_a \pi(a|s)\, P(s'|s,a)$ and ρ_0 the initial state distribution.

At the optimum, $\nabla_\theta G = 0$, so $\bar{\phi}_\pi = \bar{\phi}_{\text{data}}$. The gradient vanishes when the policy's feature expectations match the data.

Equivalence to NFXP. The conditional log-likelihood of the DDC logit model is $\ell(\theta) = \sum_i \log \pi_\theta(a_i|s_i)$. Its gradient is

$$\nabla_\theta \ell(\theta) = \frac{1}{\sigma}\left(\bar{\phi}_{\text{data}} - \bar{\phi}_\pi(\theta)\right), \tag{14}$$

which differs from Equation (11) only by the factor $1/\sigma$. MCE-IRL and NFXP optimize the same objective and converge to the same $\hat{\theta}$.

3.3 Theorem 3: Robustness Guarantee

The MCE distribution P_{MCE} solves a minimax problem over the set $\mathcal{C}$ of distributions matching the feature constraints.

Statement. Among all distributions in $\mathcal{C}$, the MCE distribution minimizes the worst-case KL divergence to the true (unknown) distribution P':

$$P_{\text{MCE}} = \arg\min_{P \in \mathcal{C}} \max_{P' \in \mathcal{C}} D_{\text{KL}}(P' \| P). \tag{15}$$

Proof sketch. The KL divergence $D_{\text{KL}}(P'\|P) = \mathbb{E}_{P'}[\log P'/P]$ is minimized (in P) when P places as much probability mass as possible on all outcomes consistent with the constraints. The maximum entropy distribution does exactly this: it is the "flattest" distribution in $\mathcal{C}$, making it the centroid of the constraint set in the KL geometry. Any other distribution $P \in \mathcal{C}$ concentrates mass away from some region, creating a worst-case P' that concentrates mass precisely in that region, yielding higher KL divergence. For a formal proof, see Chapter 3 of Ziebart (2010).

3.4 Identification

Rewards are identified only up to potential-based shaping (Ng et al., 1999). For any $h : \mathcal{S} \to \mathbb{R}$, the shaped reward $r'(s,a) = r(s,a) + h(s) - \beta \sum_{s'} P(s'|s,a) h(s')$ produces the same optimal policy (Magnac and Thesmar, 2002). The parametric restriction $r = \theta^\top \phi$ with $K < |\mathcal{S}|(|\mathcal{A}| - 1)$ resolves this by constraining the reward to a low-dimensional subspace. MCE-IRL additionally constrains $\|\theta\|$ via the feature matching dual, selecting the maximum-entropy-compatible reward from the equivalence class (Kim et al., 2021).

4 Estimation Algorithm

Algorithm 1: MCE-IRL (Ziebart, 2010)

Input: Panel $\{(s_i, a_i)\}_{i=1}^N$, features $\phi(s, a)$, transitions $P(s'|s, a)$, discount β, scale σ

`// Step 1: Compute empirical feature expectations`
$\bar{\phi}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \phi(s_i, a_i)$
Compute empirical state occupancy $D_{\text{data}}(s) = \frac{1}{N} \sum_i \mathbf{1}[s_i = s]$
Compute initial state distribution ρ_0 from first observations

`// Step 2: Initialize reward weights`
$\theta_0 \leftarrow \mathbf{0}$

`// Step 3: Iterate until convergence`
for $k = 0, 1, \ldots$ **do**

`// 3a. Backward pass: soft value iteration (Eqs. 7--9)`
$R \leftarrow \theta_k^\top \phi$ `// reward matrix` $(|\mathcal{S}|, |\mathcal{A}|)$
Solve $V = T_\sigma(V; R)$ via hybrid iteration to get V, Q, π

`// 3b. Forward pass: occupancy measure (Eq. 13)`
$F_\pi[s, s'] = \sum_a \pi(a|s)\, P(s'|s, a)$
$D_\pi = (I - \beta\, F_\pi^\top)^{-1}\, \rho_0$

`// 3c. Expected features under current policy`
$\bar{\phi}_\pi = \sum_s D_\pi(s) \sum_a \pi(a|s)\, \phi(s, a)$

`// 3d. Feature matching gradient (Eq. 11)`
$g = \bar{\phi}_{\text{data}} - \bar{\phi}_\pi$

`// 3e. Update` θ `via L-BFGS-B or Adam`
$\theta_{k+1} \leftarrow \theta_k + \alpha\, g$ `// or L-BFGS-B step`

`// 3f. Dual stopping criteria (Gleave & Toyer 2022)`
if $\|g\| < \varepsilon_1$ ***or*** $\|D_{data} - D_\pi\|_\infty < \varepsilon_2$ **then break**

end

`// Step 4: Inference`
$\ell(\hat{\theta}) = \sum_i \log \pi_{\hat{\theta}}(a_i|s_i)$
Bootstrap SEs by resampling individuals and re-estimating

Output: $\hat{\theta}$ (reward weights), V^*, π^*, log-likelihood, bootstrap SEs

5 Results

5.1 Generated Known-Truth Results

This section is generated by `mce_irl_run.py` from the `experiments.known_truth` harness. The estimator is the low-level `MCEIRLEstimator` with `optimizer=root` and `compute_se=False`. Hard gates are enforced on feature matching, occupancy moments, normalized reward/value/Q recovery, policy distance, and Type A/B/C counterfactual regret.

Table 2: MCE-IRL known-truth validation cells.

Cell	Role	States	Actions	Reward features	Individuals	Periods
Canonical low-dimensional	sanity	21	3	4	2,000	80
Low-state high-reward-feature	primary	25	3	8	3,000	100

Table 3: MCE-IRL recovery metrics.

Metric	Sanity cell	Primary cell
Feature residual	0.000000	0.000000
Occupancy moment residual	0.000814	0.001060
Reward normalized RMSE	0.007252	0.082287
Policy TV	0.004529	0.006984
Value normalized RMSE	0.007398	0.082646
Q normalized RMSE	0.026314	0.081560
Type A regret	0.000126	0.000433
Type B regret	0.000516	0.000410
Type C regret	0.000139	0.000094

Table 4: Hard-gate audit by cell.

Cell	Passed gates	Failed gates
`canonical_low_action`	10	0
`mce_low_high_reward`	10	0

The JSON artifact `mce_irl_results.json` contains the full parameter vectors, empirical and expected feature moments, occupancy residuals, raw RMSE values, normalized RMSE values, and all gate records. Raw parameter cosine is intentionally not a validation gate for MCE-IRL.

The validation uses the shared known-truth DGP rather than a real-data example. The DGP supplies the true reward, transition law, value function, policy, occupancy measures, and Type A/B/C counterfactual oracles. The success claim is therefore structural recovery under known transitions and known action-dependent reward features, not a high likelihood value by itself.

Reward, value, and Q recovery are reported after the standard IRL location-and-scale normalization. The policy and counterfactual gates are not normalized: they test whether the recovered reward actually induces the right behavior in the baseline and intervention environments. Raw parameter cosine is not used as a gate for MCE-IRL.

Code. The companion script `mce_irl_run.py` in this folder generates the JSON and LaTeX artifacts. The MCE-IRL estimator is implemented in `src/econirl/estimation/mce_irl.py`. To regenerate:

```
PYTHONPATH=src:. python papers/econirl_package/primers/mce_irl/mce_irl_run.py --enforce-gates
```

References

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